
Immersit: Evaluating Interaction Styles for 3D Venue Previews in Event Platforms

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Abstract

Online event ticketing often lacks clarity, with 2D maps failing to provide a realistic sense of seat location and view. This pilot study explores the use of 3D venue previews and compares two interaction styles (swiping and tilting) to determine which better supports users in making confident and informed seat choices. In an A/B test with 15 participants, results showed a slight preference for swiping over tilting. Swiping was associated with a higher sense of informedness, while tilting slightly increased users' confidence in their choice. A Chi-square test confirmed that the order in which interactions were tested did not significantly

affect the outcome. Although the findings are preliminary due to the small sample size, they provide valuable insights into designing more intuitive and supportive seat selection experiences. Future research should include a larger, more diverse user base to further validate these results.

Author Keywords

3D seat selection; Mobile interaction design; Mobile UX in ticketing; Venue visualization; Touch interaction; Swipe navigation; Tilt navigation; Gyroscope interaction; Sensor-based interaction; Swiping vs Tilting.

Introduction

Buying event tickets often feels like a guessing game, especially when it comes to choosing the right seat. Many people struggle with not knowing what their view will be like and whether it will let them fully enjoy the moment. Standard 2D venue maps on event platforms like Ticketmaster tend to make the ticket-buying process even more confusing, as they don't give a real sense of the space, stage distance, or venue layout. This can result in people choosing seats based on instinct, which may lead to disappointment or even discourage them from purchasing a ticket at all.

By introducing a 3D visualization of the venue, users could preview the view from their seat and get a better understanding of what to expect during the real event. Recognizing the impact this could have on the decision-making process of the customers, our project aims to evaluate two suitable interaction styles for navigating this 3D visualization - swiping and gyro-based input. Following a quantitative study with participants, we suggest which of the two is more suitable for the development of such a preview. The outlined results are based on how informed and confident their choice of users feel after using each of the interactions.

Literature review

Users and ticket purchasing in 3D seat selection

One key study in the field [1] examines the integration of accelerometer and gyroscope sensors in mobile applications. Their findings highlight the potential of tilt-based and sensor-enabled interactions to create immersive and intuitive experiences. These interaction styles allow users to engage more physically and naturally with spatial content, which aligns with the goal of improving user understanding in 3D seat selection contexts.

Additionally, [2] emphasizes the value of multimodal and user-centered interaction approaches in improving the overall user experience. Their work underlines how interfaces that allow for flexible and responsive input—such as both swiping and tilting—tend to score higher in terms of usability and satisfaction. Applying this to ticket purchasing, allowing users to explore a virtual venue through varied control types may increase their sense of confidence and control in the decision-making process.

Moreover, [3] provides a methodological framework for conducting usability testing on mobile interfaces. Their framework is relevant for evaluating how users interact with different navigation modalities (e.g., swipe vs. tilt) and how this impact cognitive load, task completion time, and perceived informedness. This can help designers assess not only which control method is more intuitive, but also which one more effectively supports goal completion—namely, the confident and informed purchase of a ticket.

Technology

Joshi, Kar and Cohen [4] look at navigating panoramic images by tilting the device and using head tracking, as well as using 'Position control', which is most like our solution. A crucial difference is that only horizontal movement is permitted, since they are working with panoramic images, instead of photospheres or entire 3D environments. They state: "A user study confirmed there is no appreciable difference in efficacy between the hybrid control and either a one finger control or a position only control." [4] This user study focused on the completion time of a task - they go on to say that "it is our belief that a view-dependent interface provides many intangible benefits such as providing a better sense of "immersion" when viewing a scene. Due to the direct interplay of a gesture and change of view in a touch-based method, it is very hard to disambiguate the sense of moving around in a virtual world versus the sensation of physically moving the world."

Thus, further research into user experience is required to see what interaction method is preferred in this type of application: swiping or gyro-based input. We will focus on quantitative research in the context of a venue

preview, but our solution and results will be more broadly applicable.

Based on the identified gap, we formulated the following research question:

What type of interaction within immersive VR venue previews best supports users' sense of being informed and confident in their ticket purchasing decisions?

Additionally, we have set two sub-questions:

- Which interaction style helps users be more informed about the position of their seat?
- Which interaction style makes users feel more confident about their seat choice?

Design

Prototype and Unity transition

We used a clickable Figma prototype to simulate the user flow of buying a ticket in Ticketmaster. For the seat selection process, we designed an interactive 3D map of the venue using Blender and Unity. A smooth transition between the two was created through mobile deep linking. This allowed users to move from Figma to Unity through specific links. This helped to avoid back-and-forth jumping between the two apps. This approach made it easier for users to explore multiple seats within the same session, with Figma only used to start and end the testing process.

Design of Unity app

Once in Unity, users began with a top-down view of the venue and could click on sections to zoom in. Available

seats were highlighted in green. Selecting a seat changed the camera to a first-person view from that position, helping users see their view before taking the decision to buy a ticket. The final selection brought the users back to Figma to complete the purchase.

Interaction types

The first interaction method allows for greater immersion: the mobile phone's gyroscope and accelerometer are used to correspond with the view in the 3D environment. This helps to create an immersive experience, where the rotation of the virtual camera is the same as the phone's rotation. We will refer to this interaction method as 'gyro-based'. A disadvantage of this interaction method is that users need to physically spin in place. Along with this immersive interaction method, a more traditional method is compared. Swiping on the screen to move the camera, as seen in Google Street View, or any panoramic 360° image viewer.

Methodology

Participants

Due to the project's time, this pilot study was performed with 15-20 young adults between the ages of 20 and 30 who have previous experience with purchasing event tickets. To recruit participants, a combination of convenience and snowball sampling was used. Additionally, the open testing day organized by the Sticky association at Utrecht University was also used to find more participants, mainly fellow students.

Materials

To collect and analyze the data of this study, the following materials were used:

- Clickable Figma prototype of Ticketmaster and Unity app with the two interaction styles
- A/B testing protocol with tasks for the participant
- Post-test 5-point Likert scale questionnaire with open questions to gather qualitative insights

Experiment design and procedure

For this pilot study, a task-based A/B test was conducted using a within-subject design, where all participants went through both interaction flows:

- Flow A - Purchasing a concert ticket using the *swiping* interaction for seat selection
- Flow B: Purchasing a concert ticket using the *gyroscope* interaction for seat selection

This study design consisted of one categorical independent variable (type of interaction) and two continuous dependent variables - level of user informedness and confidence in seat choice. Additionally, the participant's preference for interaction style was also recorded through open-ended questions.

Each user was randomly assigned to an interaction style. The participants were provided with the two versions of the Ticketmaster flow already installed on the researcher's mobile phone. In a controlled environment, the users performed the following tasks while being observed:

- Choosing a concert and opening the 3D venue visualization
- Exploring the venue (seats, stage and layout)

- Selecting a seat based on personal preference

The same tasks were performed for both interaction styles.

After the testing, participants filled in the survey with questions about level of informedness, confidence in choice and preferences for interaction style. The questionnaire was based on 3 UX scales System Usability Scale (SUS), User Engagement Scale (UEG), and Igroup Presence Scale (IPG). Questions about interaction preference and overall impressions were also included.

Results and Discussion

To evaluate whether the order of interaction that was tested had any effect on the results, a Chi-square test of independence was run for the order of interaction (swiping vs tilting) and the overall preference of the user. The test showed that there is no significant dependence between the two factors ($\chi^2 = 2.109$, $p = 0.348$) and that the final result is unaffected by the order of testing.

A paired sample t-test was run on part of the data to evaluate which interaction type had more effect on the level of informedness and confidence in seat choice. The results show that despite a low and non-significant difference, the swiping interaction helps users feel more informed about where their seat is positioned (see Appendix A.1), while the tilting contributes more to the perceived confidence in the right seat choice (see Appendix A.2).

More than half of the users prefer swiping interaction (53%) compared to tilting (40%) (see Fig.1). They

found the touch “intuitive and easy to control”. This preference might be influenced by the commonly used swipe interactions in similar applications (e.g., Google Maps Street View).

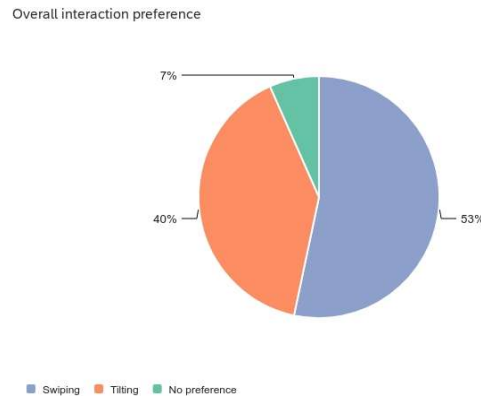


Fig. 1. Interaction Preference between Swiping and Tilting

On the other hand, while the tilting felt “immersive, informative and more realistic”, users pointed out feeling “dizzy after the usage of the gyro functionality” and stating that such option will be unnatural to use in specific context, such as crowded spaces or on a train.

Some suggestions for improvements that were received from users include an introductory screen with control instructions to minimize confusion, pricing details for each seat, and the option to view a crowd.

For the future, participants are overall more likely to use a 3D preview when it has a swiping control, compared to tilting. Although the gyro-based interaction also received high results, there was some

polarization, leading to the conclusion that tilting may be suitable for a specific user group and context (see Appendix A.4).

Limitations and future work

While this study provides insights into which interaction style is suitable for 3D previews, it has limitations. First, the small sample size may affect the generalizability of the findings. A larger and more diverse sample could reveal broader preferences across different user groups with varying levels of technical proficiency. Second, the evaluation focused only on short-term impressions. This limits our understanding of how interaction methods influence satisfaction after purchase. Long-term metrics such as correlation between satisfaction during and after the event should be assessed. This could offer deeper insights into the value of immersive visualization tools. Lastly, as users may interact with event platforms on both mobile and desktop, future research could compare interaction strategies across platforms to develop consistent and flexible user experiences.

Conclusion

This study examined which interaction style (swiping vs. tilting) in 3D venue previews in event platforms made users more informed about their seat position and confident about their choice. An A/B testing with 15 participants revealed that with very small differences, the majority of users prefer touch-based control compared to gesture based. Results show that swiping increases informedness levels; tilting contributes more to perceived confidence. Due to the short-term focus and small number of participants, future research should expand on this topic to evaluate the long-term satisfaction of a larger and more diverse user group.

References

- [1] Caro-Alvaro, S., Garcia-Lopez, E., Brun-Guajardo, A., Garcia-Cabot, A., & Mavri, A. (2024). Gesture-based interactions: Integrating accelerometer and gyroscope sensors in the use of mobile apps. *Sensors*, 24(3), 1004.
<https://doi.org/10.3390/s24031004>
- [2] Lee, S., & Kim, K. (2023). Human-computer interaction: Enhancing user experience in interactive systems. *E3S Web of Conferences*, 399, 04037.
<https://doi.org/10.1051/e3sconf/202339904037>
- [3] Garcia, L., & Chen, H. (2024). Usability testing of mobile applications: A methodological framework. *Applied Sciences*, 14(5), 1792.
<https://doi.org/10.3390/app14051792>
- [4] Joshi, Neel & Kar, Abhishek & Cohen, Michael. (2012) Looking at you: Fused gyro and face tracking for viewing large imagery on mobile devices. Conference on Human Factors in Computing Systems - Proceedings.
<http://doi.org/10.1145/2207676.2208375>

Appendix

Appendix A: Graphs

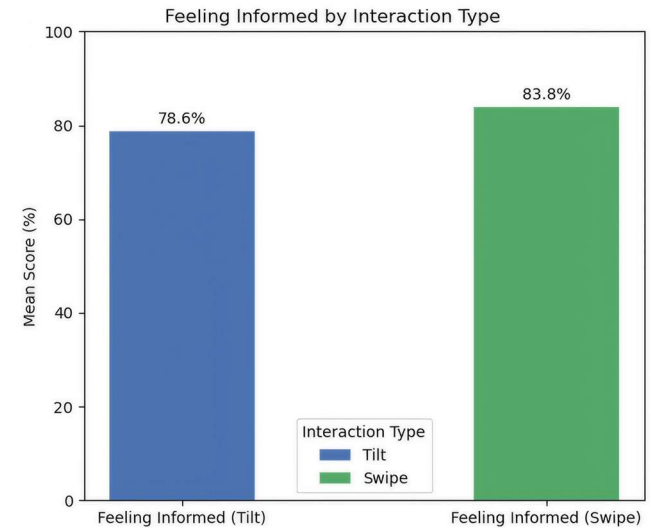


Fig. A1. Comparison over Feeling Informed by the specific Interaction Type

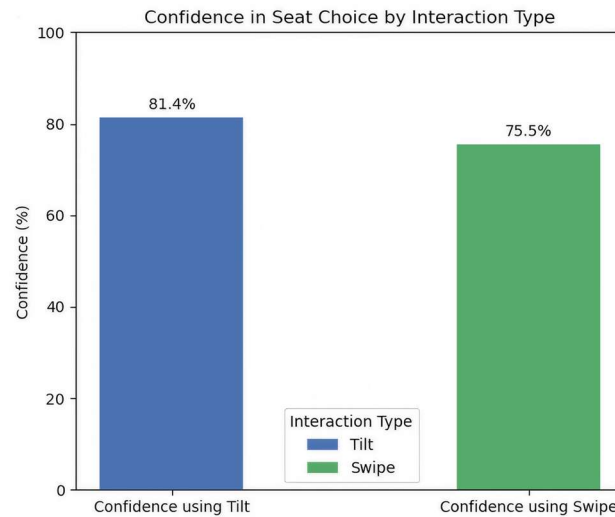


Fig. A2. Comparison of the Confidence in Seat Choice by the specific Interaction Type

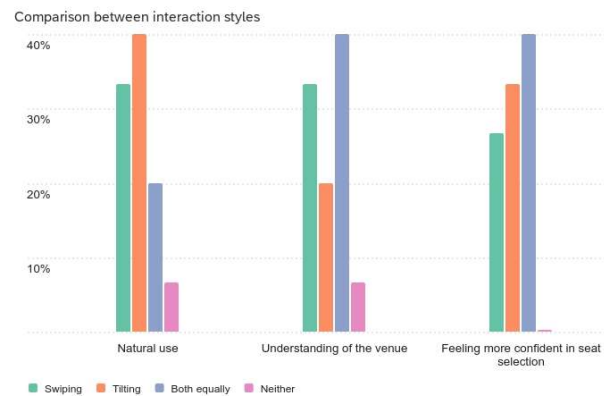


Fig. A3. Overall Comparison Between the Interaction

style, based on: Natural Use, Understanding of the venue, Feeling More Confident in Seat Selection

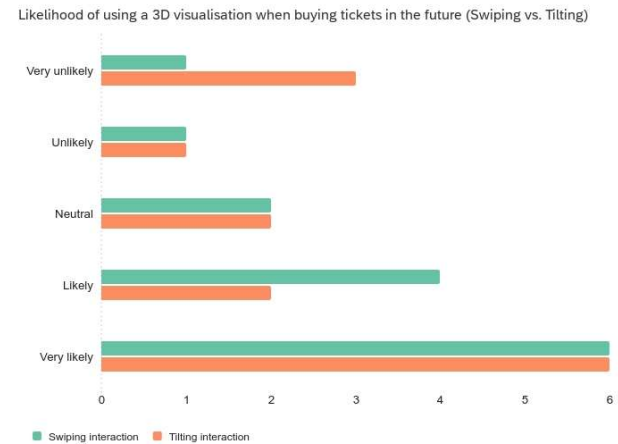


Fig. A4. Comparison of the Likelihood of Using a 3D visualization by specific Interaction Type When buying Tickets in the Future

Appendix B: Consent Form

Thank you for considering taking part in this testing session conducted by team "Immersit" from Utrecht University. The purpose of this study is to evaluate the design and functionalities of the digital prototype we have created for the "Mobile Interaction" class. Before you decide whether to participate, it is important that you understand the purpose of the study and what your participation will involve. Please read the following information carefully.

What will this testing involve?

If you agree to participate, you will be asked to:

Complete a ticket-purchasing flow using two different interaction prototypes (swipe-based and tilt-based)

Exploring the digital venue

Selecting/changing a seat based on their personal preferences

Identifying areas that are either crowded or more accessible

Complete a short questionnaire about your experience after testing

Additional testing aspects:

Your interaction with the mobile prototype will be observed by one of the members of the group

The order of the two prototypes will be randomly assigned

The duration of the testing session will be around 15-20 minutes

You may choose to skip any questions in the survey questionnaire or tasks you are uncomfortable with

In this session, the prototype and interactions are being tested, not your actions or choices

Voluntary Participation

Your participation is completely voluntary. You may withdraw at any time without providing a reason. If you choose to withdraw, any data collected up to that point may be used unless you request its deletion.

Confidentiality and Data Protection

Your responses will be kept confidential and only used for the purpose of this study. All data collected will be anonymized and stored securely before analysis. Only the members of the team will have access to the data. All collected data will be deleted within 1 month after analysis.

Your input from this testing will help our group to further improve the design, usability, and interactions of the final product.

Questions and Contact Information

If you have any questions about this study, you may contact i.i.dimitrova@students.uu.nl.

Consent Statement

By signing below, you acknowledge that you have read and understood the information above and that you voluntarily agree to participate in this study.

Thank you for participating!

Participant Name:

Student email

Appendix C: Survey Questionnaire

Interaction order

(SELECT WHICH ONE WAS FIRST)

- Swiping
- Gyroscope

Comparison & Overall Impressions

In the following section you will be asked some questions regarding your overall experience with both interactions.

Which interaction felt more natural to use?

- Swiping
- Tilting
- Both equally
- Neither

Which interaction gave you a better understanding of the venue layout?

- Swiping
- Tilting
- Both equally
- Neither

Which method helped you feel more confident in selecting a suitable seat?

- Swiping
- Tilting
- Both equally
- Neither

Which interaction did you prefer overall?

- Swiping
- Tilting
- No preference

How likely would you like to use a system like this (3D venue with such interactions) when purchasing

tickets in the future? (Likert Scale 0 – 5, for both Swiping Interaction / Tilting Interaction)

- 0 - Would never use
- 1 - Extremely unlikely
- 2 - Somewhat unlikely
- 3 - Neither likely nor unlikely
- 4 - Somewhat likely
- 5 - Extremely likely

Final Thoughts

Did anything about either interaction confuse you or slow you down? If so, please describe it.

Was there anything missing from the interaction that you expected?

What suggestions do you have for improving the 3D venue experience?

Any other thoughts you'd like to share?

Appendix D: Prototypes

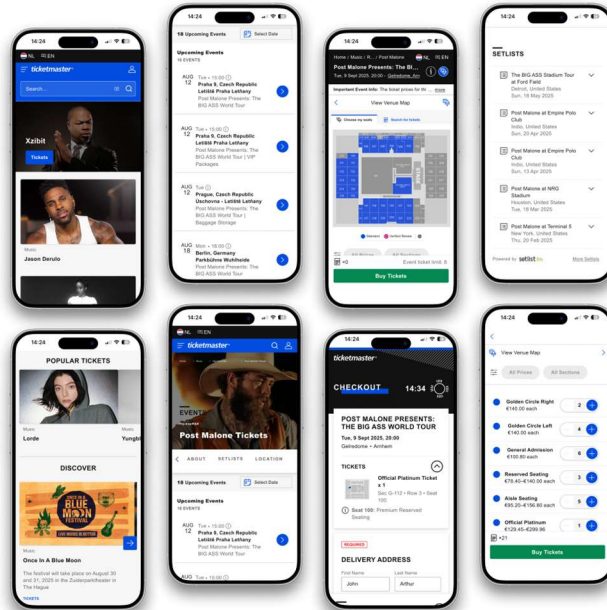


Fig. A5. Figma Prototype, capturing the storytelling elements of our project



Fig. A6. Unity Prototype, capturing the technical implementation – Portrait Mode

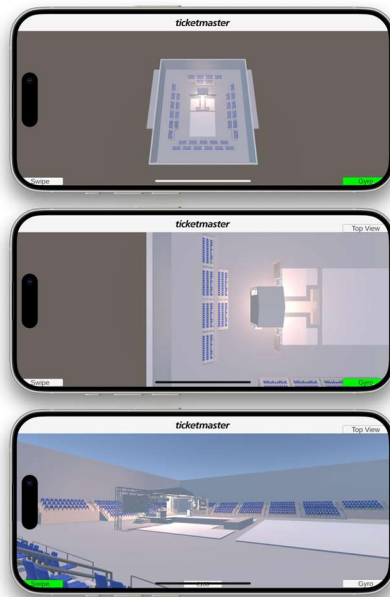


Fig. A7. Unity Prototype, capturing the technical implementation – Landscape Mode